Tashkent Institute of Textile and Light Industry
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Editor-in-Chief
IEEE Internet of Things Journal

Dear Editor-in-Chief,

We submit the manuscript *"An IoT-Enabled Hybrid Soft Sensor for Real-Time Residual-Moisture Monitoring and Resource Accounting in Compact Wool Scouring Machines"* for consideration as a **Regular Paper** in the IEEE Internet of Things Journal. We request **Traditional (non-open-access) publication**.

The paper addresses IoT trials/experiments and IoT application design, both within the scope of the journal. Its main contributions are: (i) a low-cost ESP32-based IoT architecture for wool scouring machines, which today operate without instrumentation; (ii) a hybrid soft sensor that uses a design-of-experiments model of the squeezing device as a physics-informed prior and is recalibrated online from sparse laboratory samples; (iii) real-time per-kilogram energy and water indicators and fault alarms; and (iv) a reproducible reanalysis of the historical mechanical-process baseline and an open-source implementation (https://github.com/nasin0764-bit/woolwash-iot) with a defined prospective field-validation protocol. [TODO: Update (iv) with the completed field validation before submission.]

The manuscript is original, has not been published previously, and is not under consideration elsewhere. [TODO: If any part was presented at a conference, cite it here and explain the extension.] [TODO: Insert ONLY after every author has read and approved the complete manuscript:] All authors (R. Tuguzbayeva, S. Khakimov, F. Dustova, R. Rosulov, and M. Anorkulov) have approved the submission and declare no conflicts of interest.

[TODO: Author contributions – confirm with each author before submission. Proposed for M. Anorkulov (pending his confirmation): statistical reanalysis, including verification of the factorial experiment, regression coefficients, and RMSE/MAE/R^2 calculations.]

Suggested reviewers (no conflict of interest, outside our institution):

1. [TODO: Name, affiliation, e-mail – industrial IoT / soft sensors]
2. [TODO: Name, affiliation, e-mail – textile machinery / wool processing]
3. [TODO: Name, affiliation, e-mail – edge computing for manufacturing]

Sincerely,

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on behalf of all co-authors
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